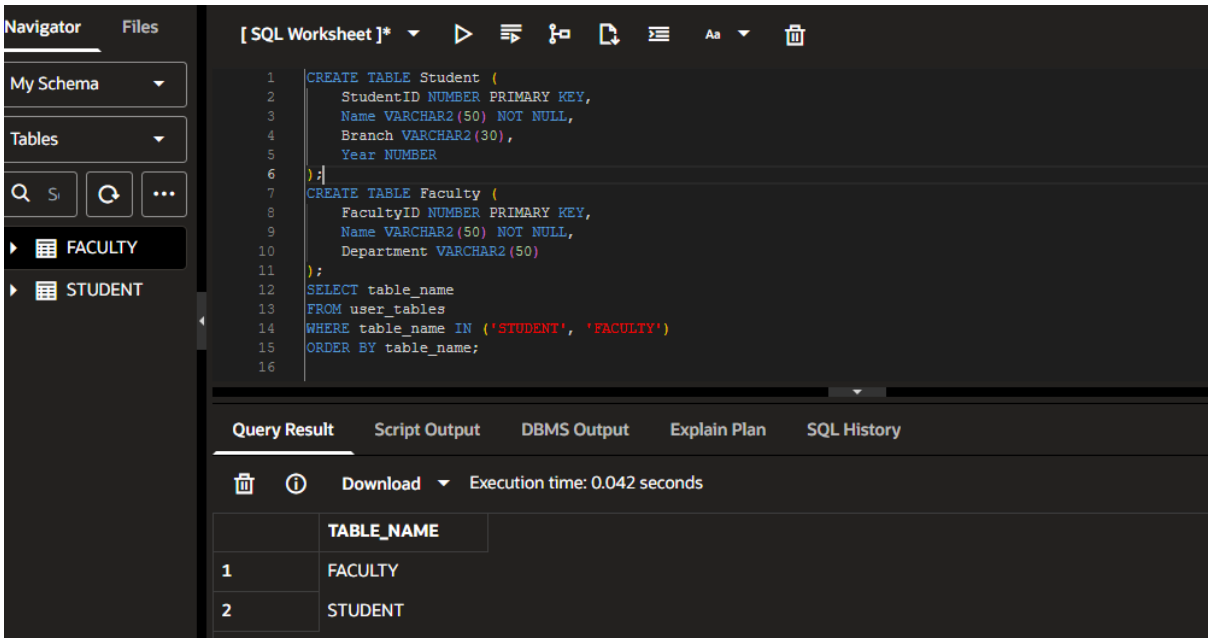
LAB ASSESSMENT 2

QUESTION 1.

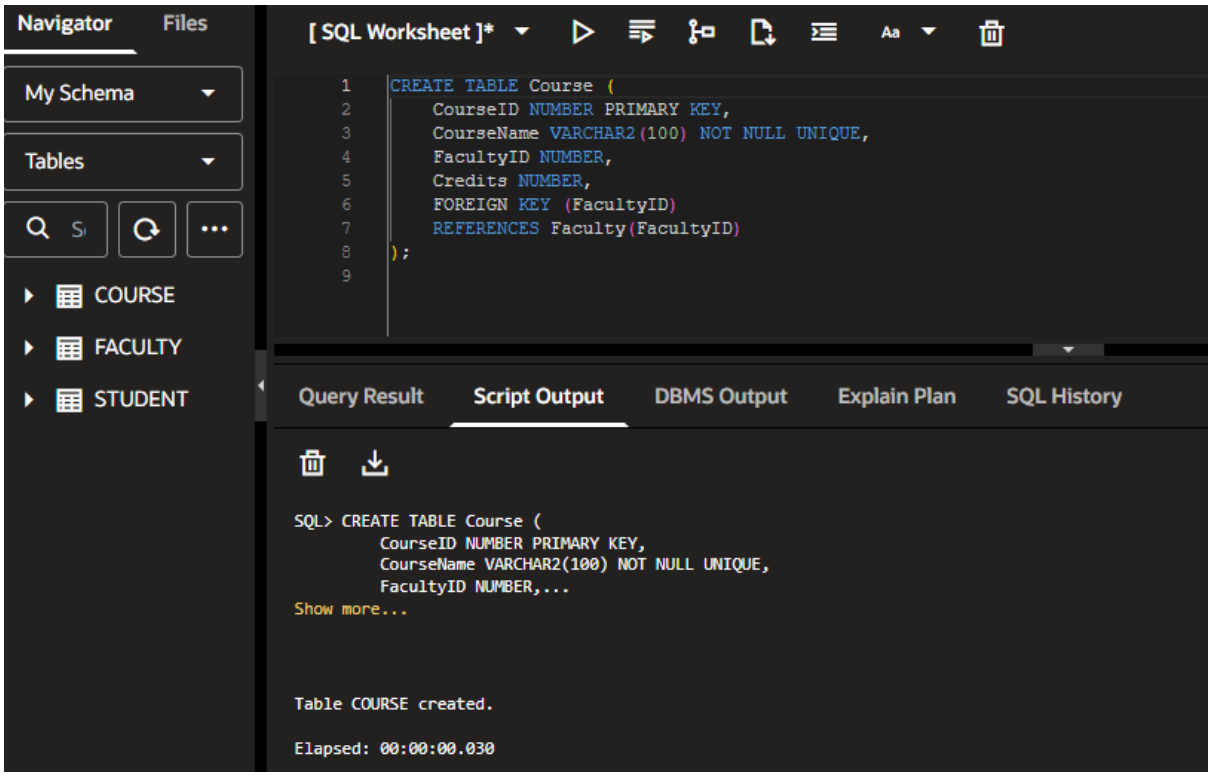
Write DDL statements to create the four tables mentioned above. Each table must have a Primary Key. `Course.FacultyID` must be a Foreign Key referencing `Faculty`, while `Exam.StudentID` and `Exam.CourseID` must be Foreign Keys referencing `Student` and `Course`, respectively.

Apply `NOT NULL` on Name fields and `UNIQUE` on an appropriate alternate column. Run the table creation statements and confirm with `DESCRIBE` output for all four tables before proceeding.

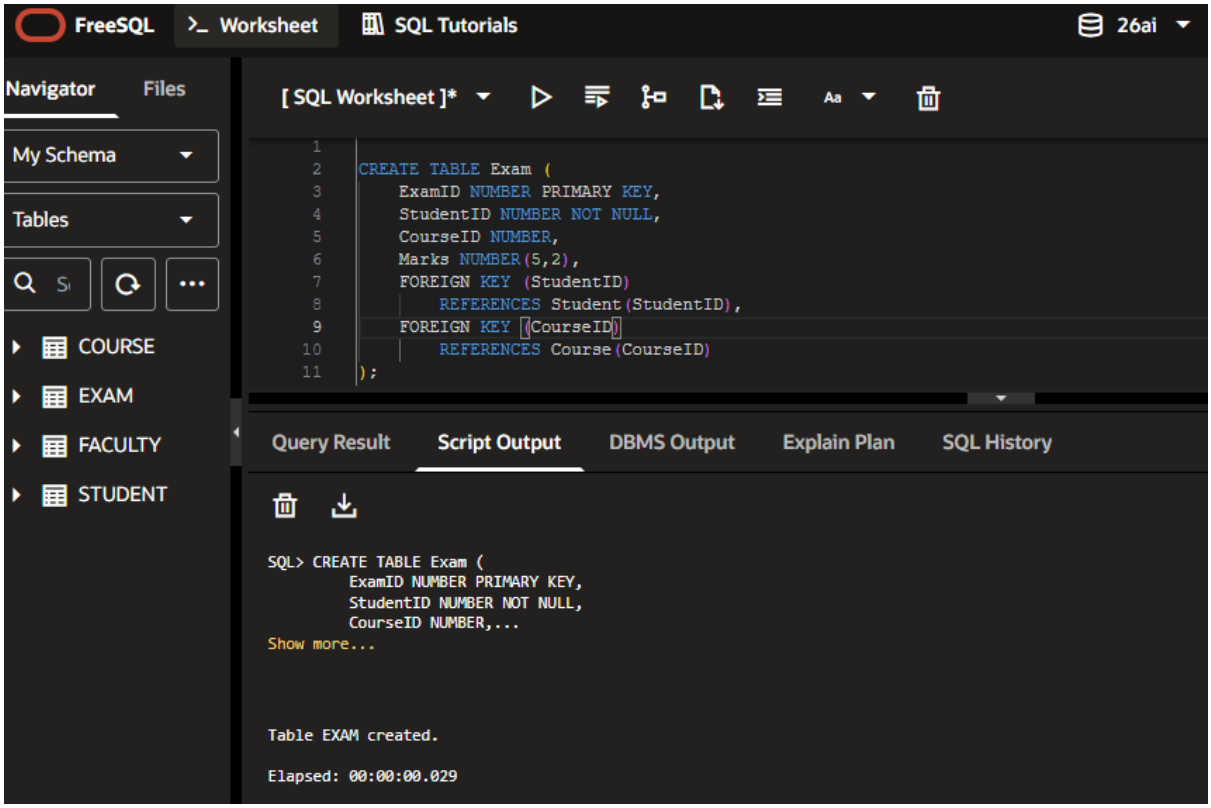
Creating the Student table using the CREATE TABLE statement and Creating Faculty table wit its primary key and NOT NULL constraints.



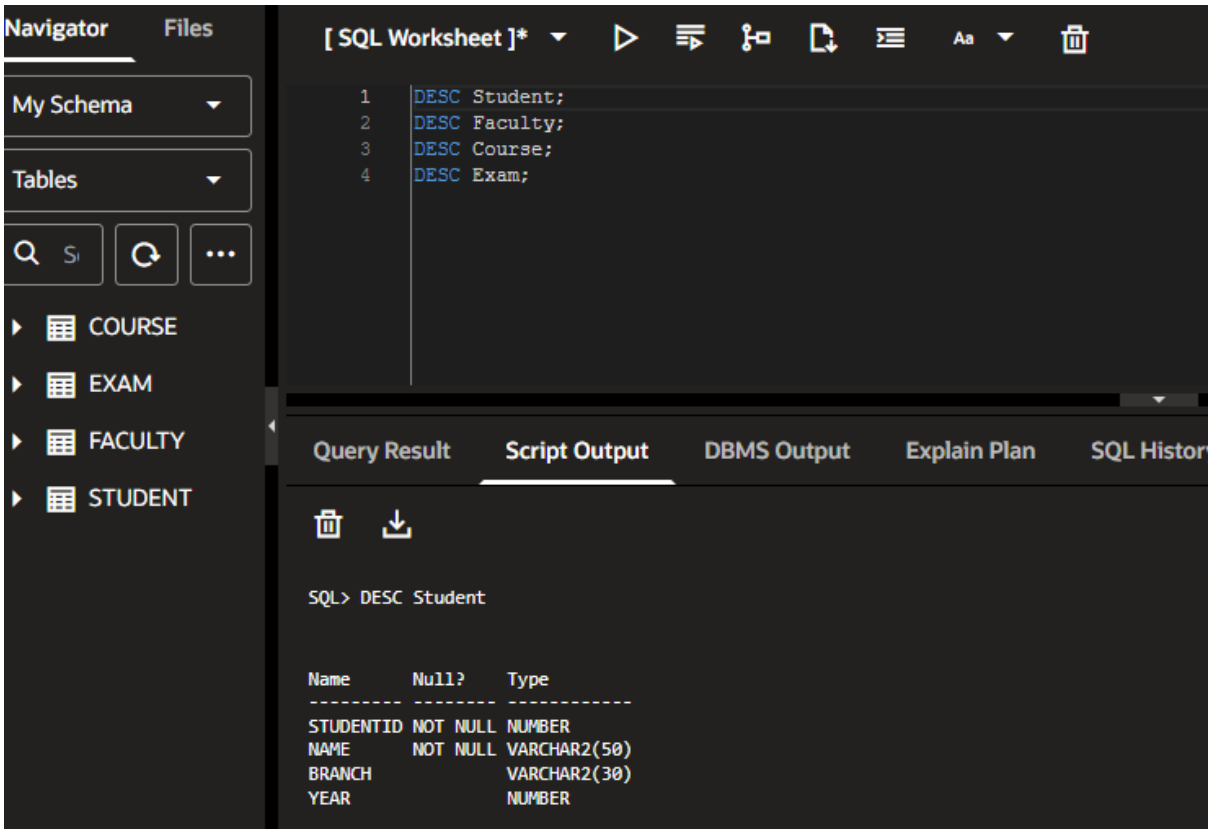
Creating the Course table with a foreign key referencing Faculty.



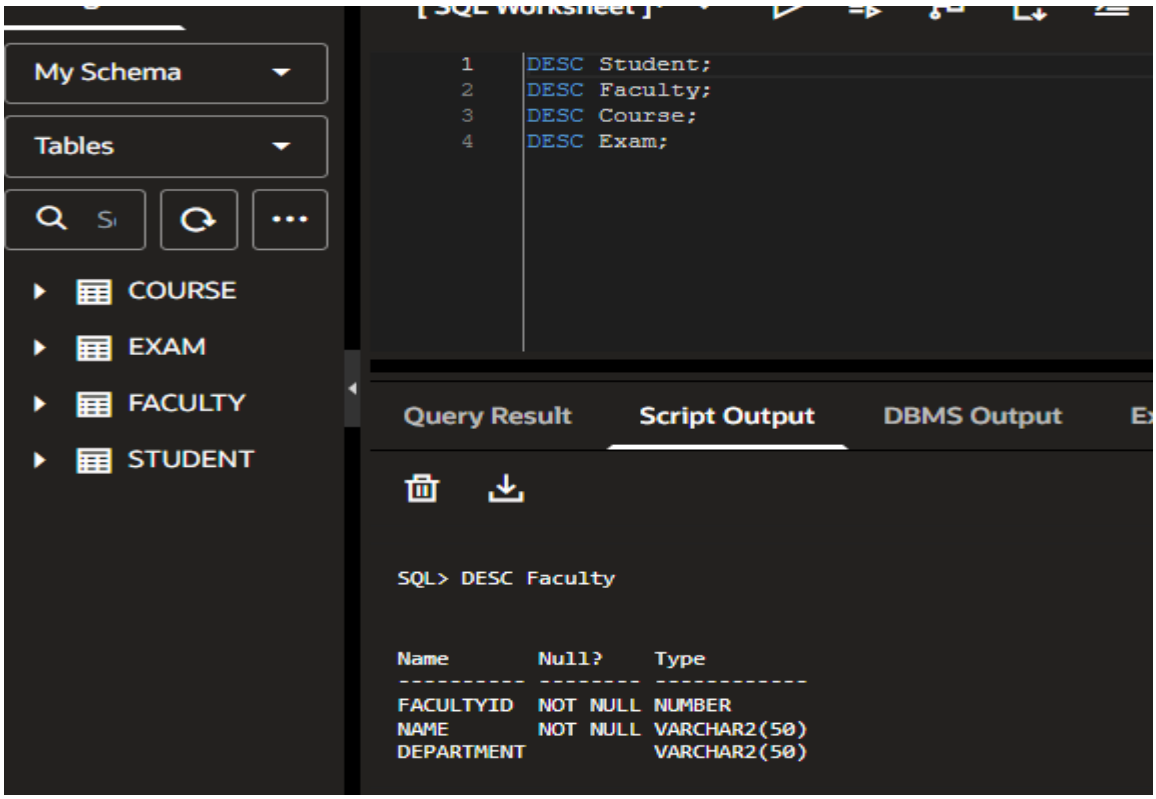
Creation of the Exam table with ExamID as the Primary Key and StudentID and CourseID as Foreign Keys referencing the Student and Course tables, respectively.



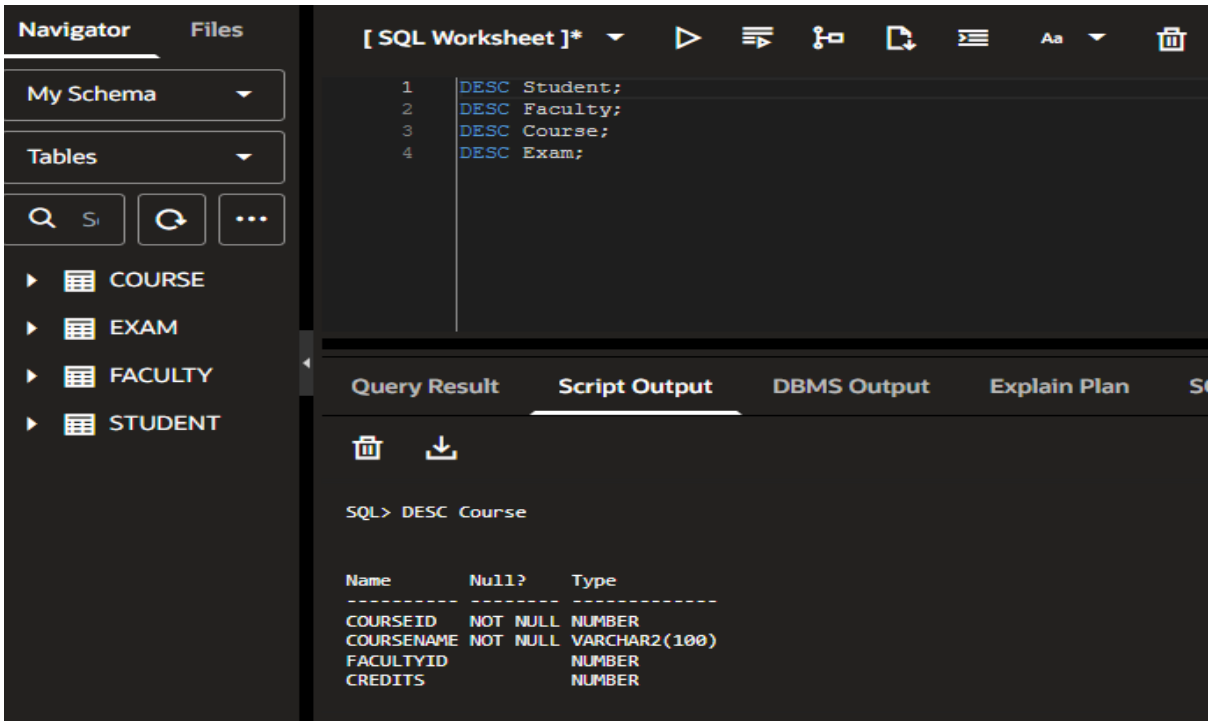
Describe Student.



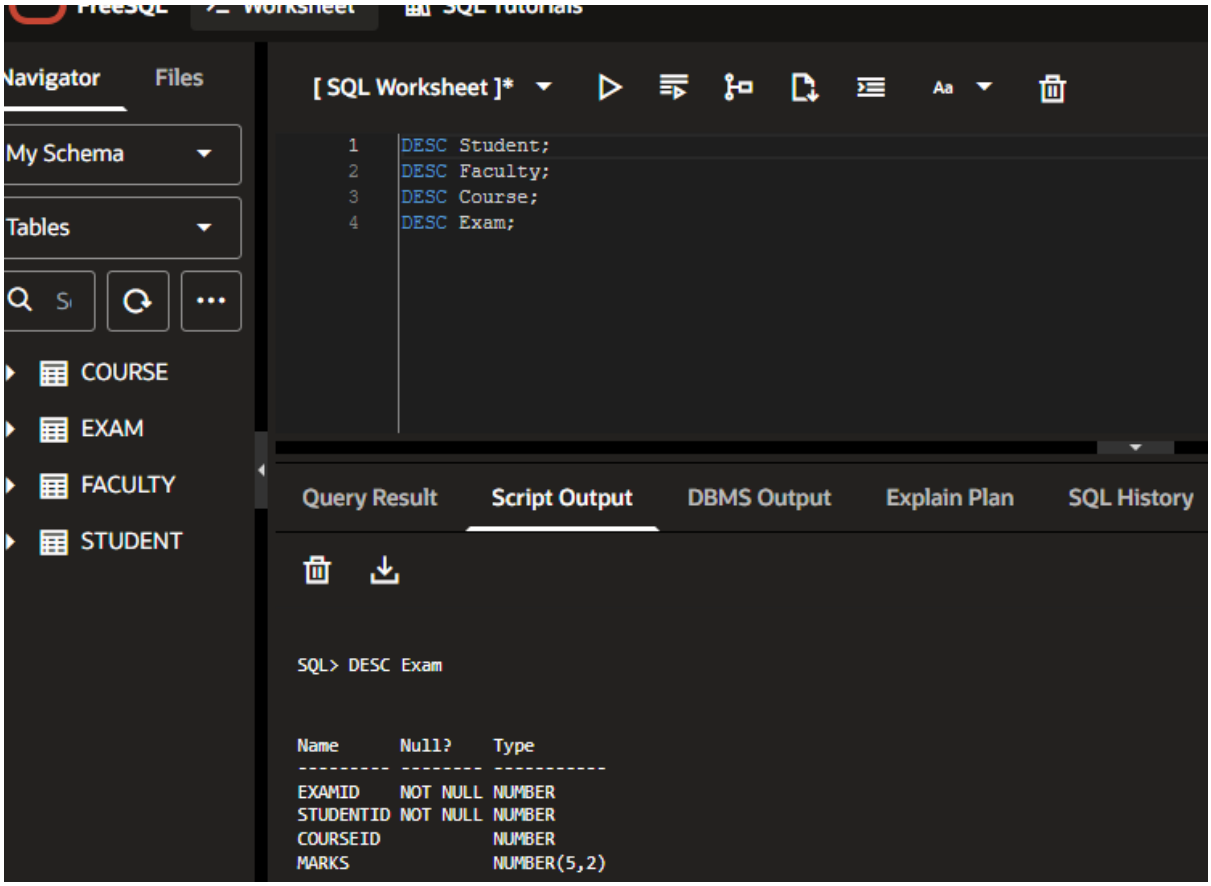
Describe faculty.



Describe Course.



Describe Exam.



QUESTION 2.

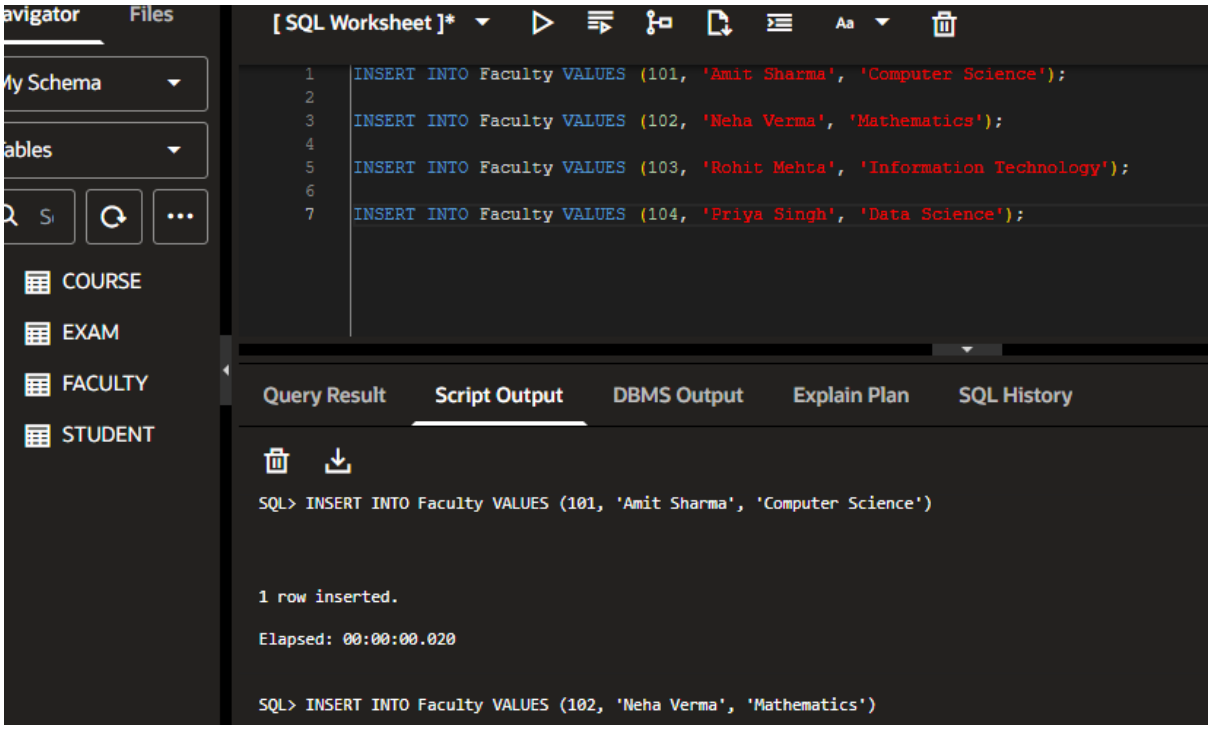
Insert at least 8 Students, 4 Faculty, 5 Courses and 12 Exam records, ensuring enough variety to support later GROUP BY and JOIN queries (e.g., multiple students per course, at least one student with no exam record yet).

Then:

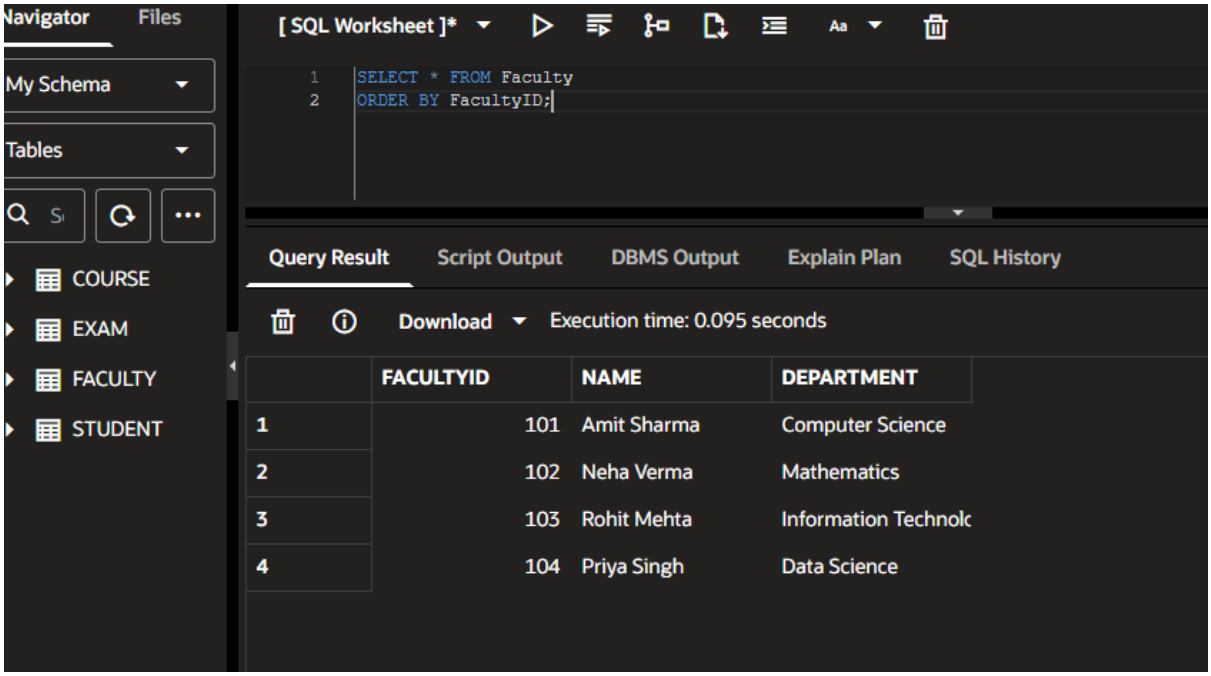
(a) UPDATE one student's Branch.

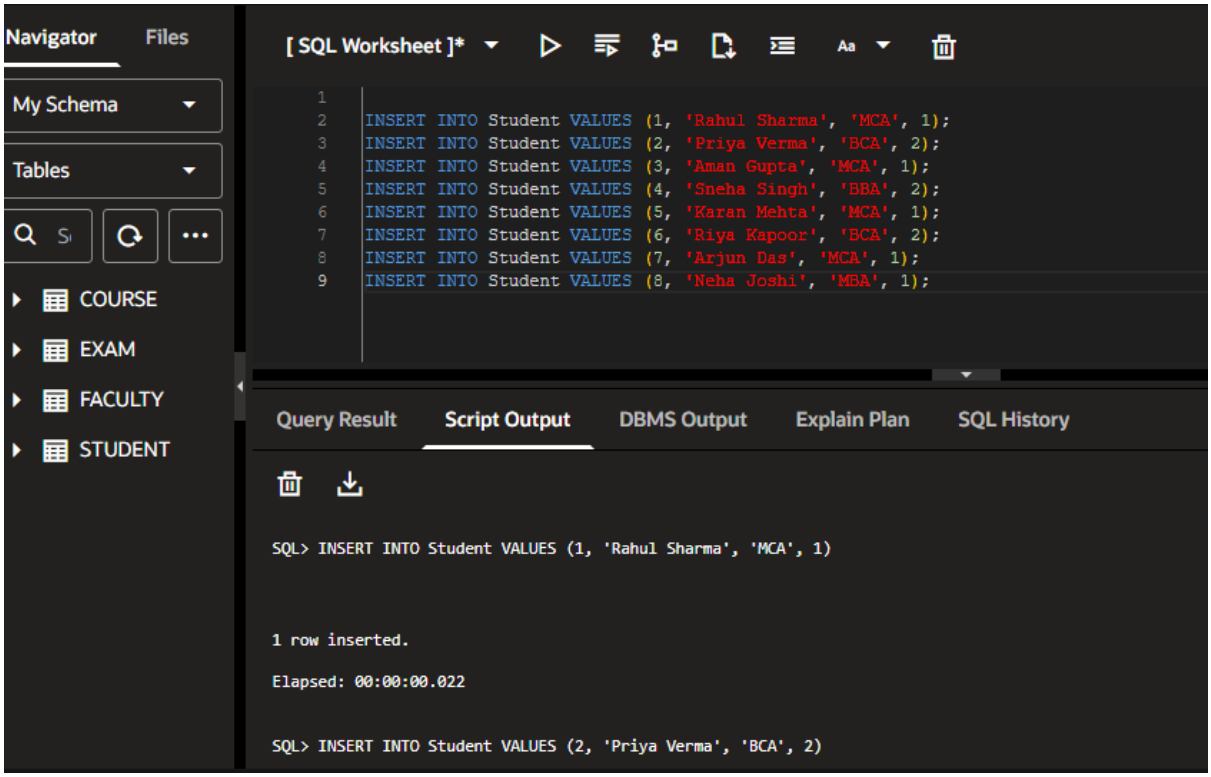
(b) DELETE one Exam record.

(c) Attempt to INSERT an Exam record with a non-existent StudentID and show/explain the resulting Foreign Key error.

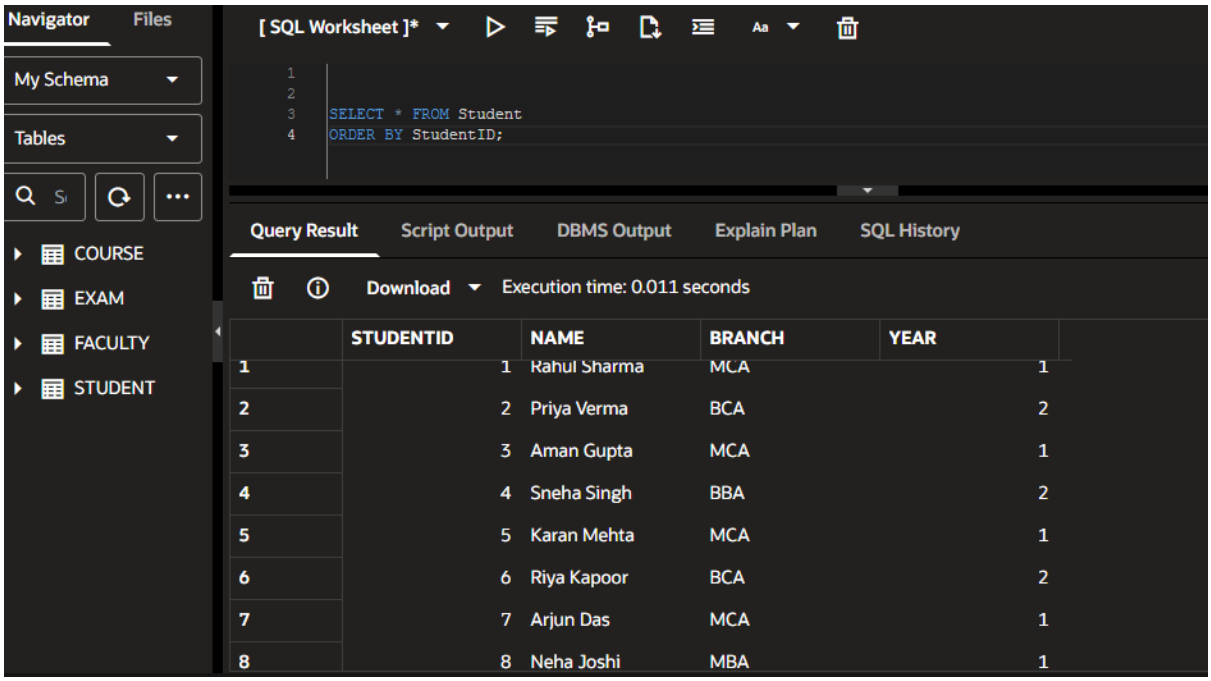


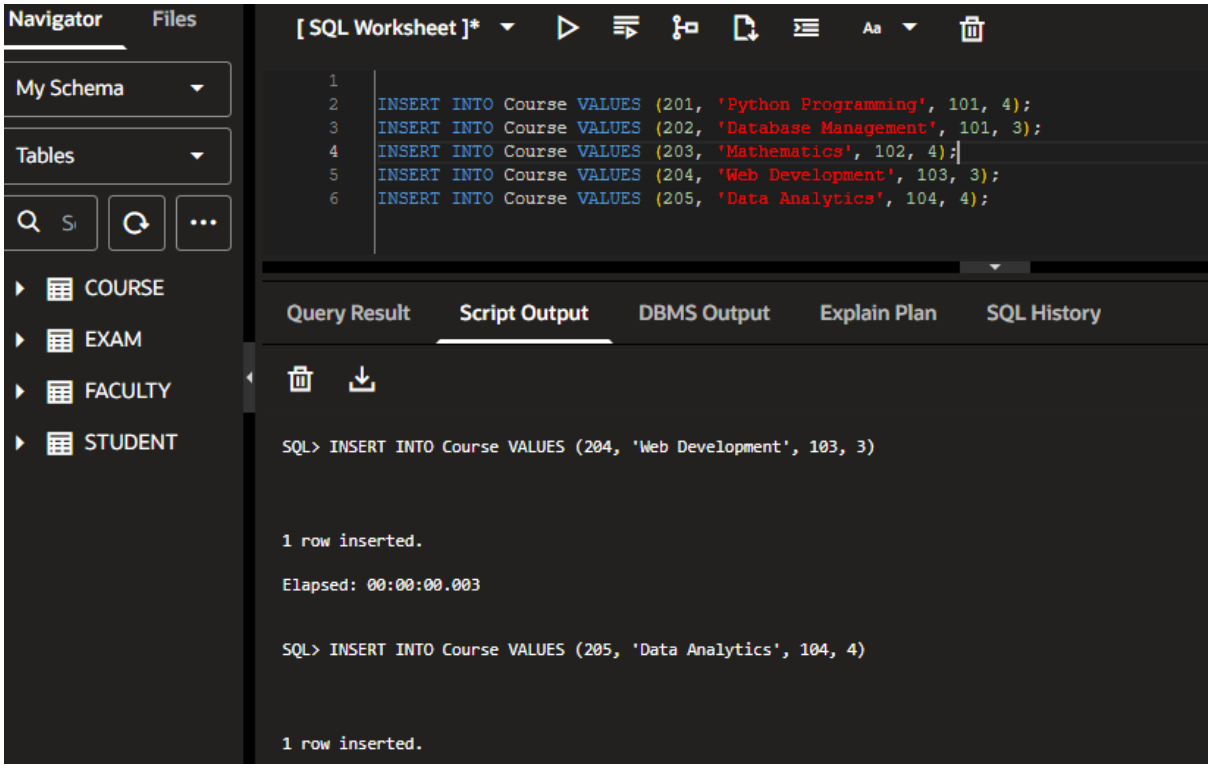
Displaying the four successfully inserted faculty records using the SELECT statement.



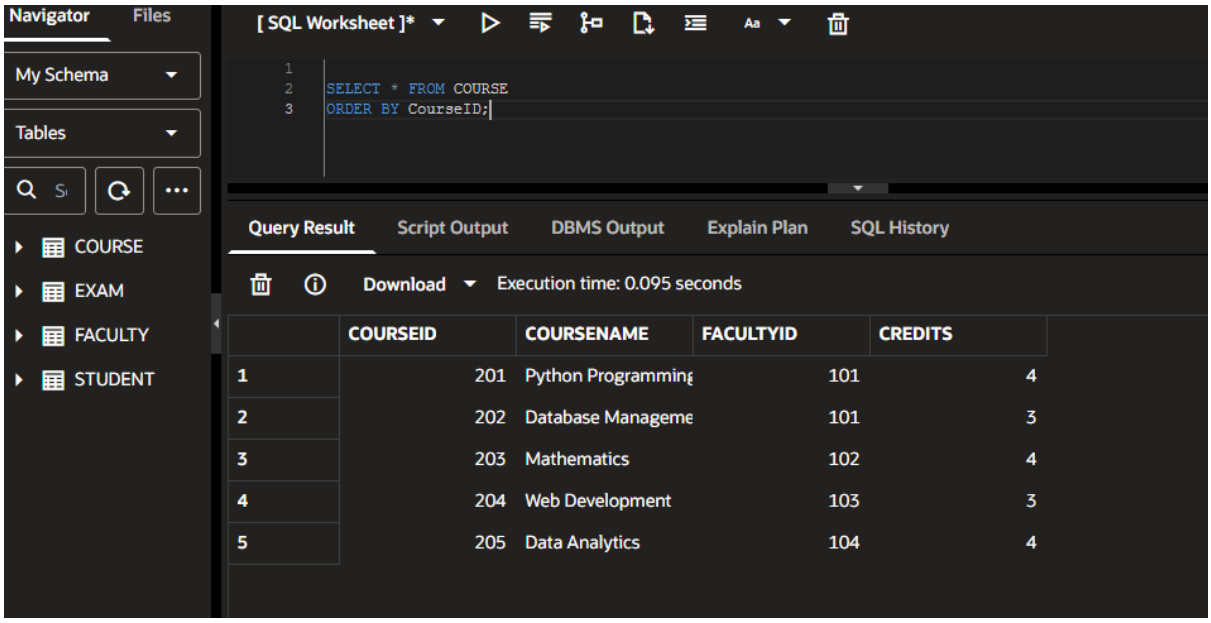


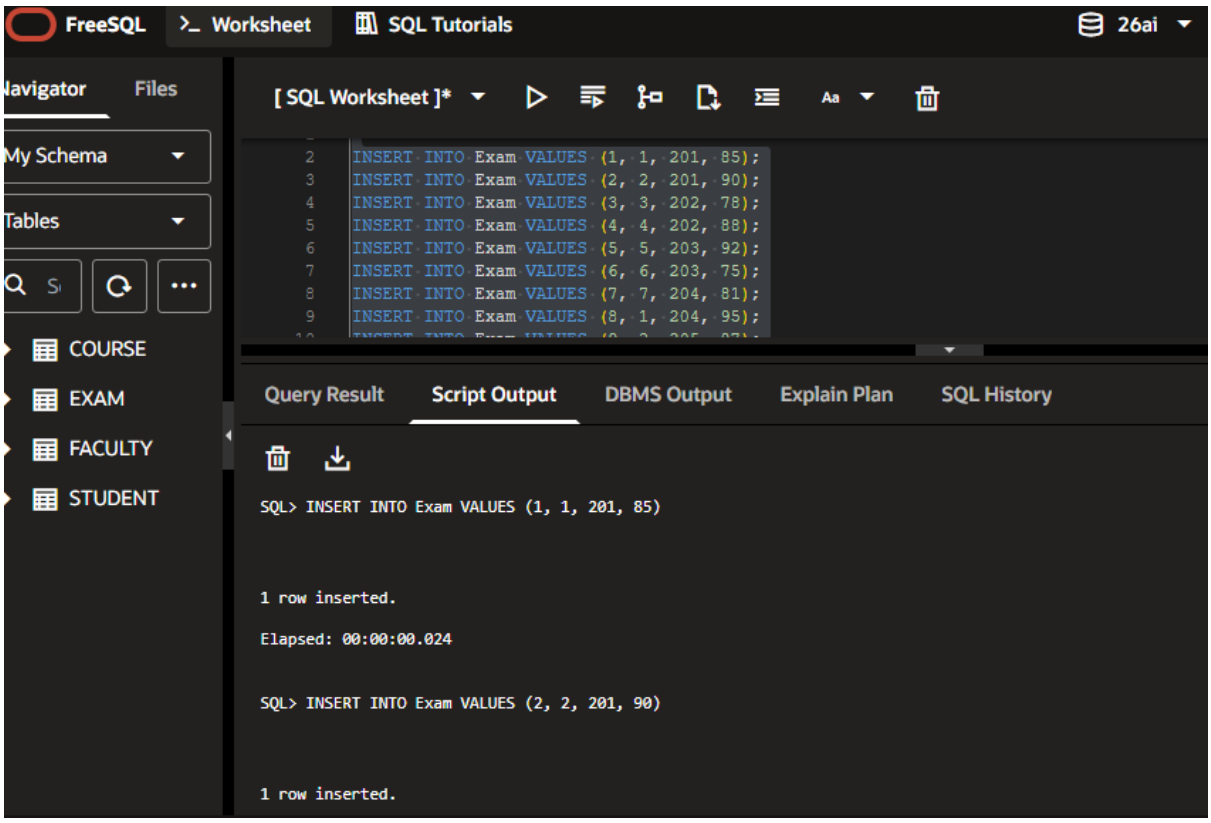
Displaying eight student records inserted into the Student table using the INSERT and SELECT statements



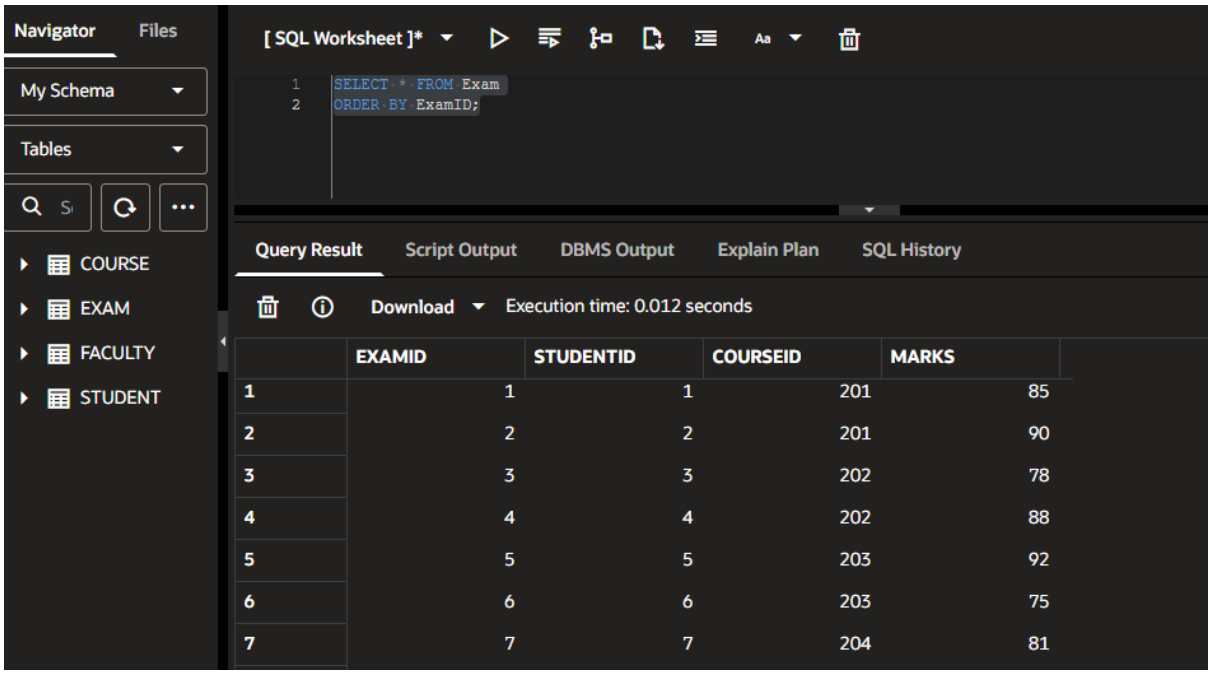


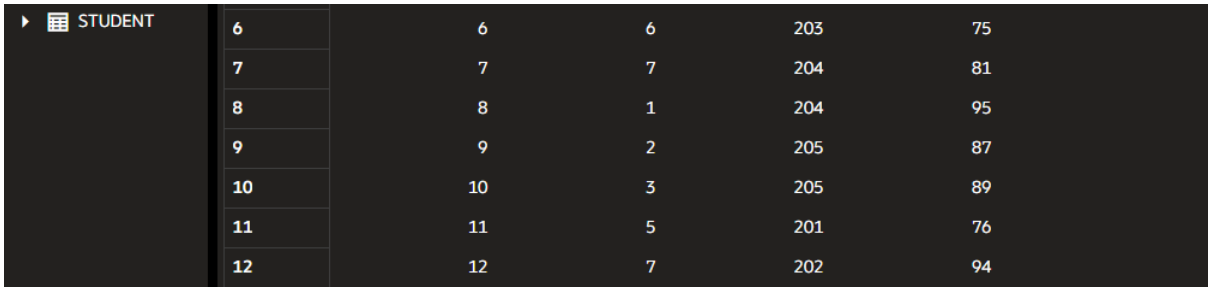
Displaying five successfully inserted course records from the Course table using the SELECT statement

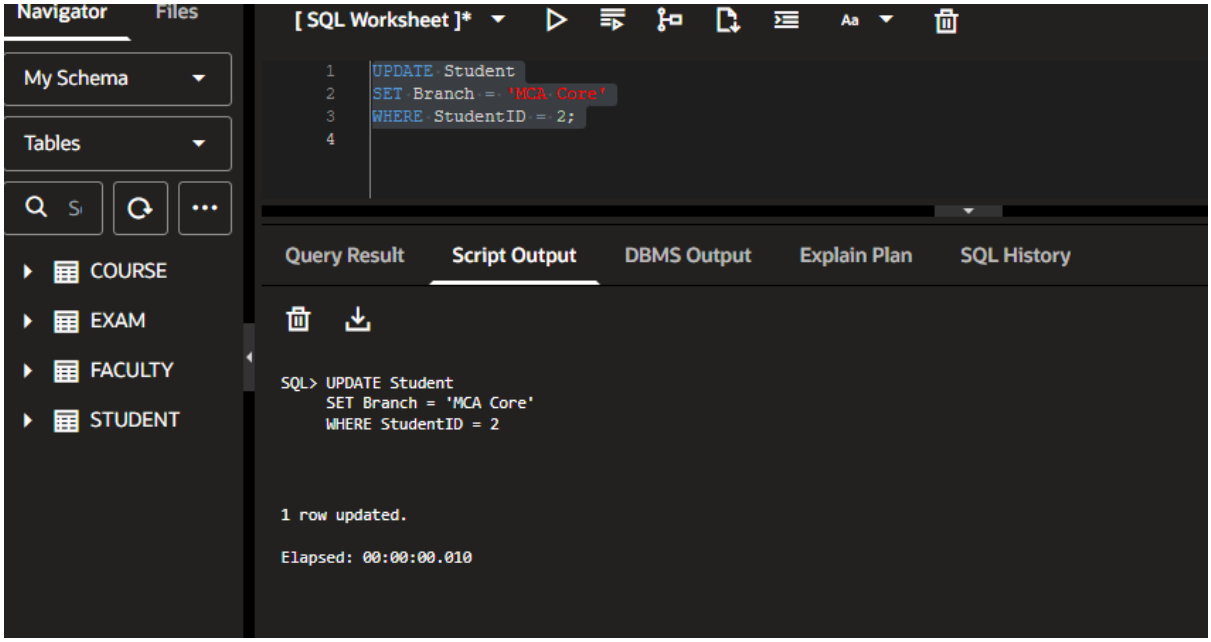




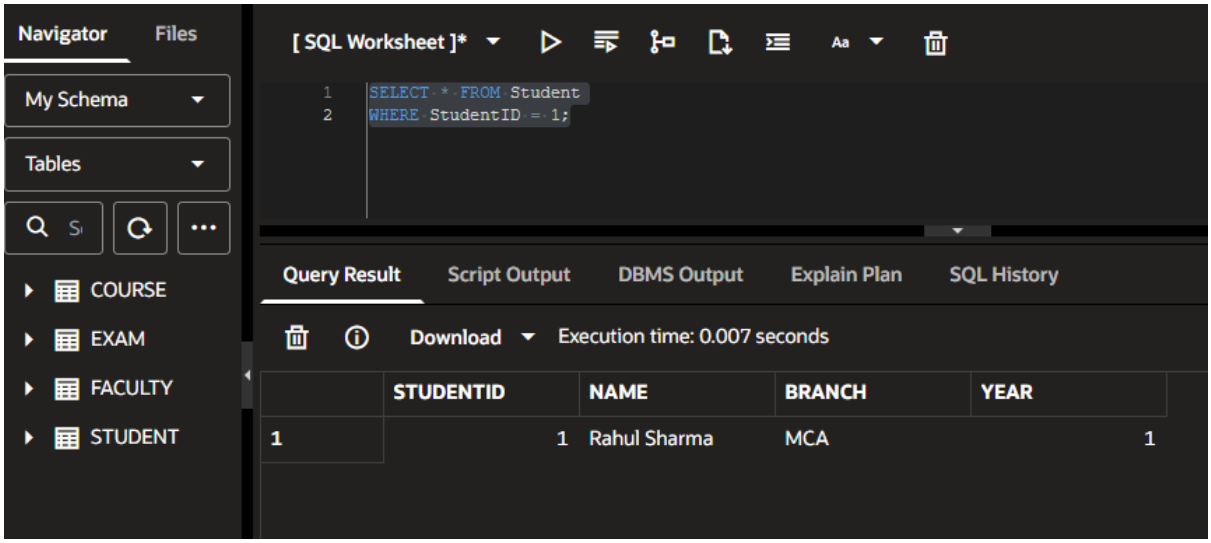
Inserting and displaying twelve examination records in the Exam table using INSERT INTO and SELECT statements.

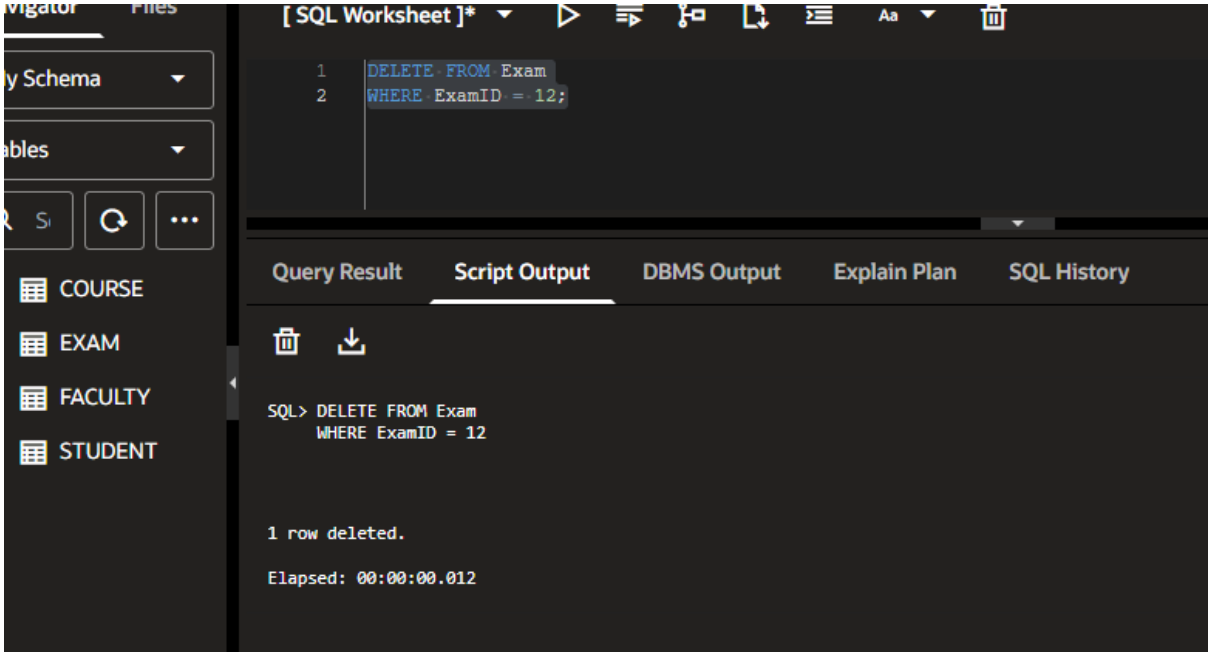




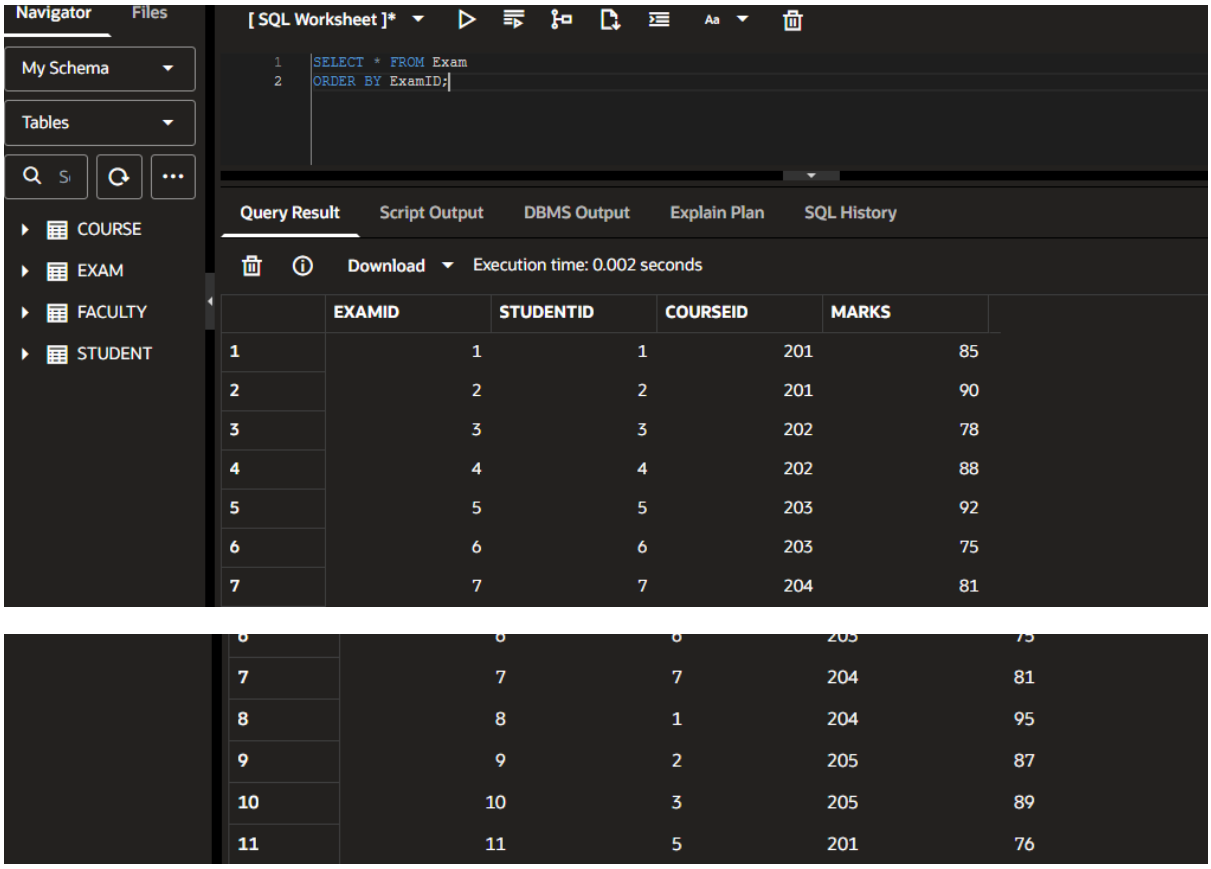


Updating the Branch of a student in the Student table using the UPDATE statement and verifying the modified record using SELECT.





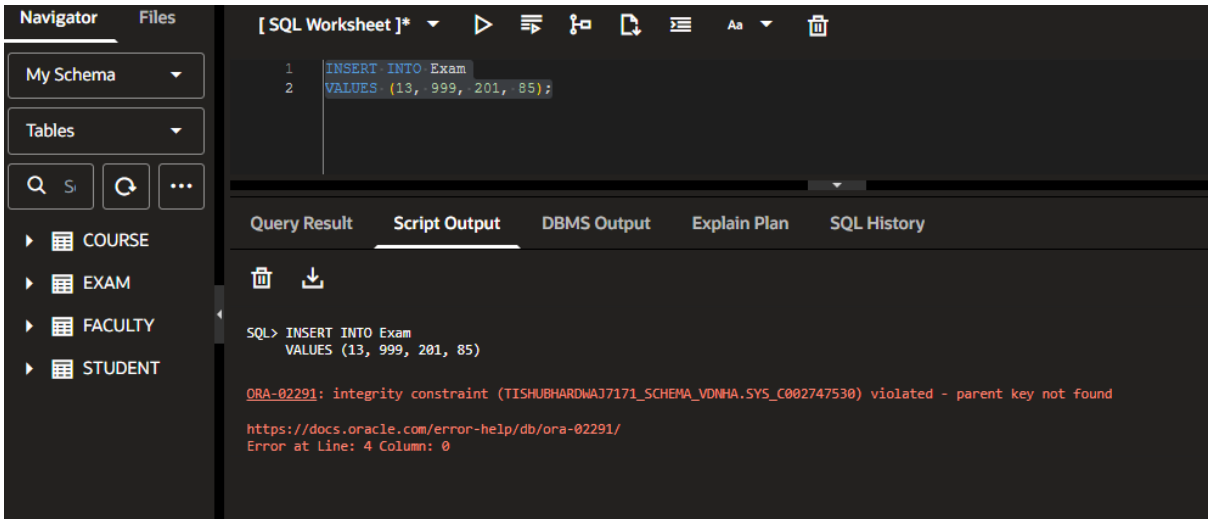
Deleting an examination record from the Exam table using the DELETE statement and verifying the updated table



	EXAMID	STUDENTID	COURSEID	MARKS
1	1	1	201	85
2	2	2	201	90
3	3	3	202	78
4	4	4	202	88
5	5	5	203	92
6	6	6	203	75
7	7	7	204	81

7	7	7	204	81
8	8	1	204	95
9	9	2	205	87
10	10	3	205	89
11	11	5	201	76

Demonstrating Foreign Key constraint violation by attempting to insert an Exam record with a non-existent StudentID (999). Oracle generates ORA-02291 because the referenced parent record is not available in the Student table.



QUESTION 3.

Q3. Implementation of WHERE Clause

(Uses Q1–Q2 data only — no new tables/data)

Write and run the following SQL queries:

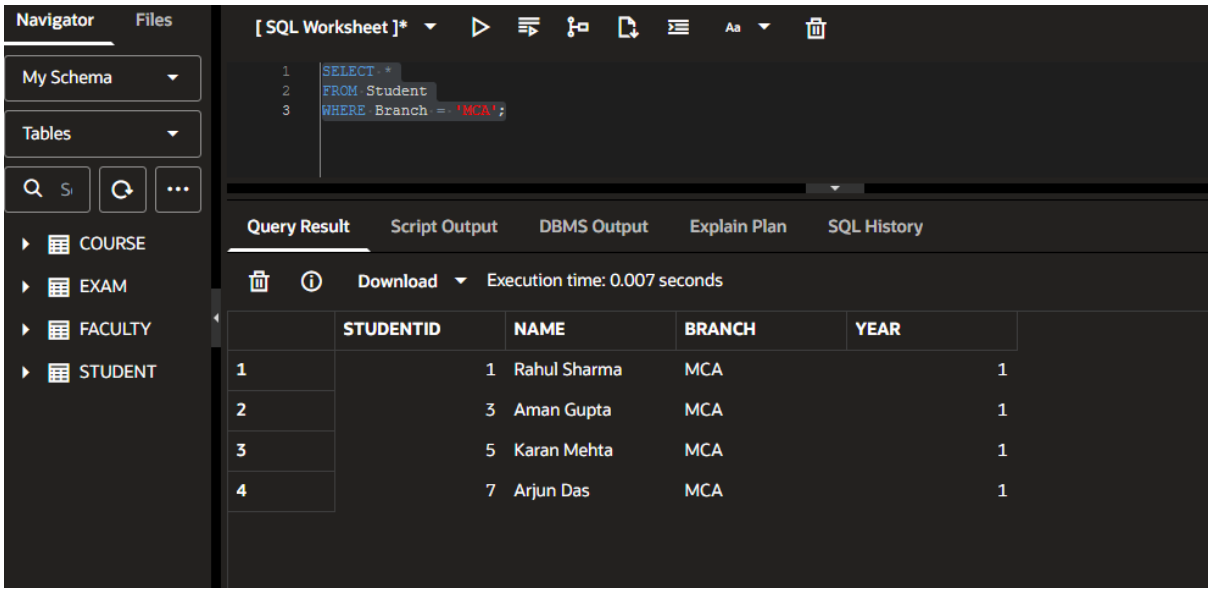
(a) Display all students from a specific Branch using the `WHERE` clause.

(b) Display all exam records sorted by Marks in descending order using `ORDER BY`.

(c) Calculate average Marks per CourseID using `GROUP BY` and `AVG()`.

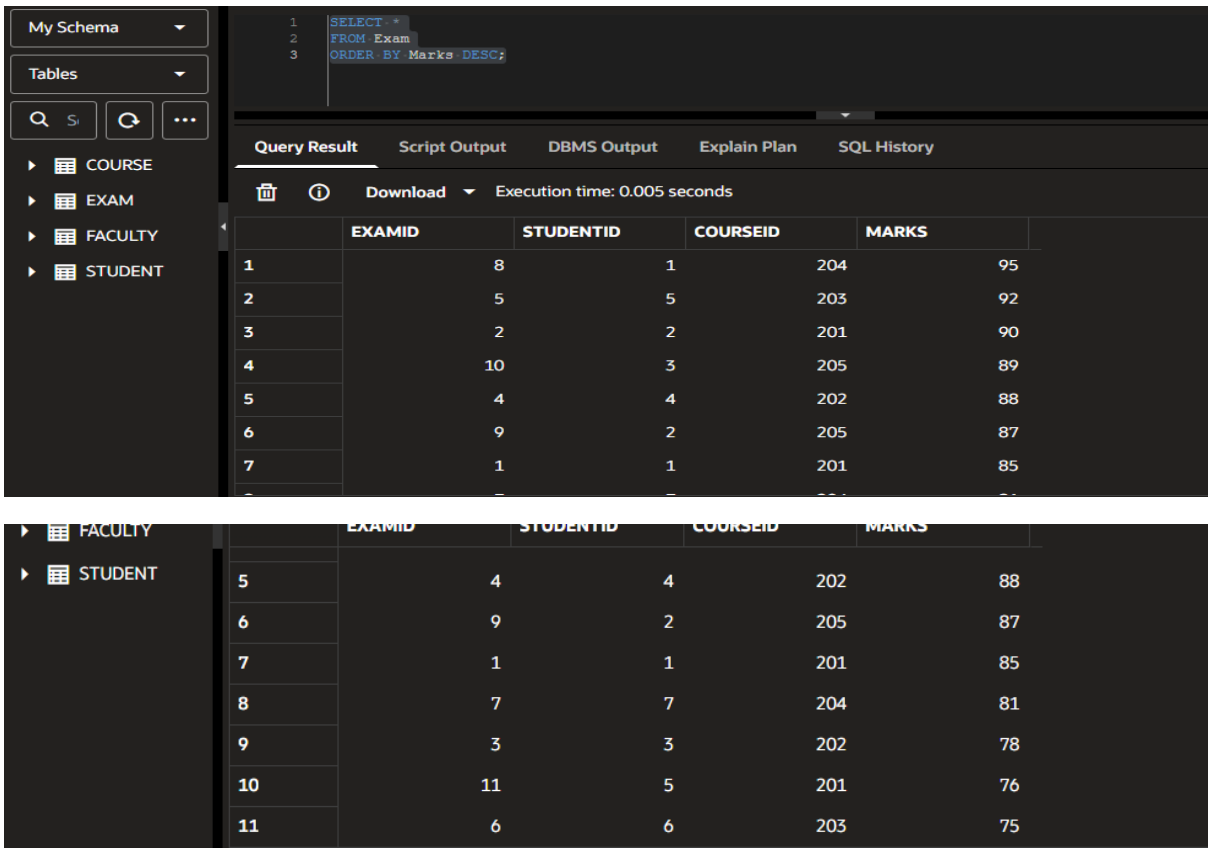
(d) Find the highest Marks scored in each Course using `GROUP BY` and `MAX()`.

a)



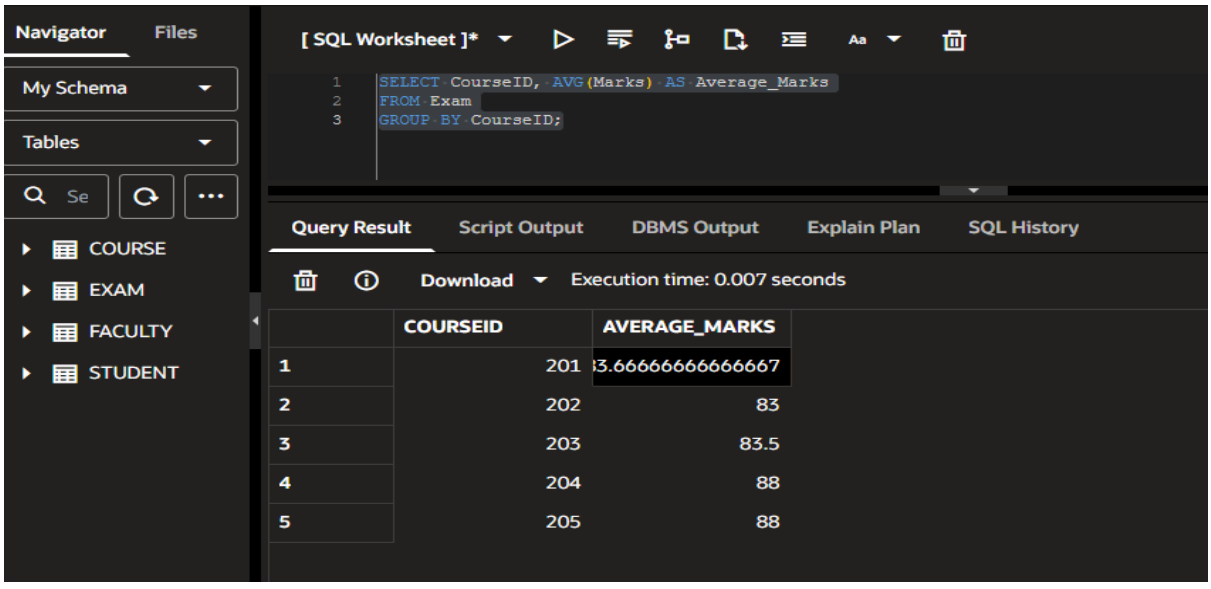
Displaying all students belonging to the MCA branch using the WHERE clause in SQL

b)



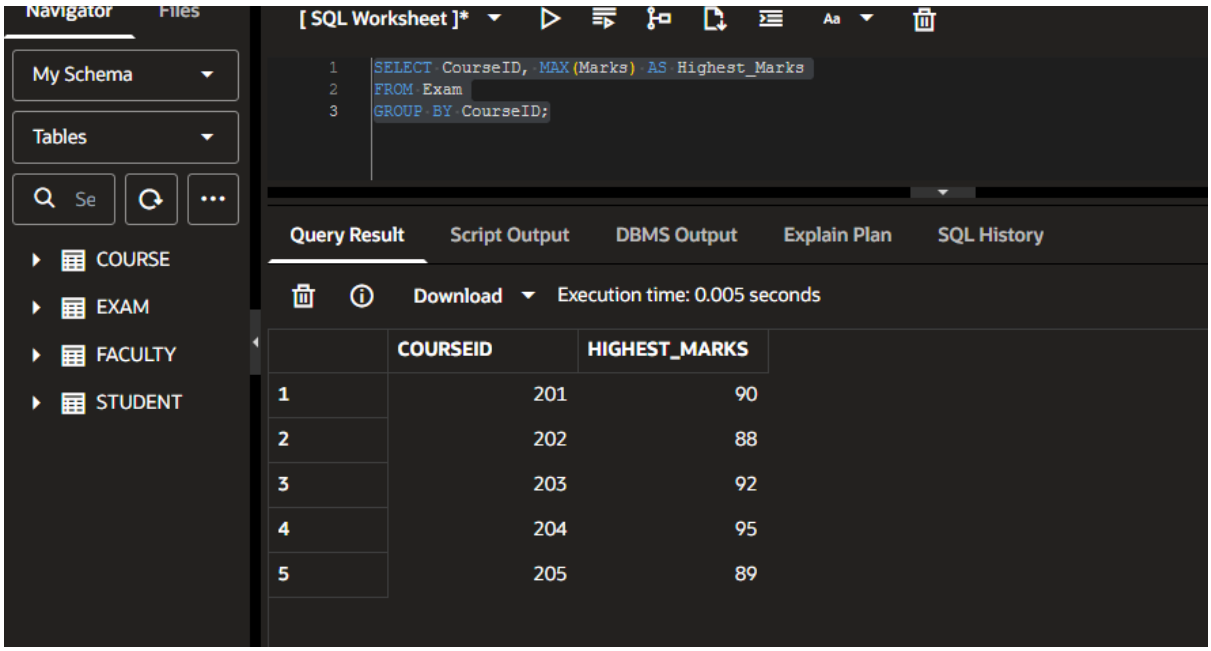
Displaying examination records in descending order of Marks using the ORDER BY clause.

c)



Calculating the average examination marks for each course using the GROUP BY clause and AVG() aggregate function.

d)



Finding the highest marks obtained in each course using the GROUP BY clause and MAX() aggregate function.